

ABSTRACT

AN AUTONOMOUS CREDIT RISK ASSESSMENT AND EXPLAINABLE AI SYSTEM WITH HYBRID MULTI-BANK POLICY RAG ENGINE

(Loan-IQ)

Loan-IQ is an intelligent, autonomous credit risk assessment and policy decision engine designed to ensure accurate, transparent, and compliant loan underwriting, especially during complex financial evaluations and multi-bank policy assessments. The system continuously monitors applicant risk profiles in real time, detects underwriting anomalies or policy violations using advanced AI/ML techniques, and automatically generates audit-ready credit scoring and decision rationale without manual intervention bottlenecks.

It combines a 2-tier Stacking Ensemble Classifier (XGBoost, LightGBM, CatBoost, Random Forest with an Artificial Neural Network Meta-Learner), SHAP explainable AI attributions, and a specialized multi-bank policy compliance module. When a loan application or policy query is submitted, Loan-IQ evaluates financial obligations (FOIR, LTV, EMI), isolates policy violations, and reroutes non-compliant requests with actionable mitigation steps. In parallel, it utilizes a Hybrid RAG Engine combining dense FAISS vector search and sparse BM25 keyword retrieval with Reciprocal Rank Fusion (RRF) re-ranking to ensure that critical bank guidelines (SBI, HDFC, ICICI, IOB, Canara, RBI) reach decision-makers accurately.

The system uses AI to predict creditworthiness, learn from historical lending data, and improve its response over time. With a user-friendly dashboard, it provides real-time alerts, credit risk visualization, and detailed logs for better decision-making. Loan-IQ aims to create a resilient, secure, and intelligent financial technology infrastructure that supports continuous, compliant credit processing in enterprise banking scenarios.

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